Last-Iterate Rates of Stochastic Fast Krasnosel'skiĭ–Mann Iterations: Momentum, Implicit Variance Reduction, and Decaying Noise

Abstract

Blue highlights the research questions and reading guide;
red highlights the main answers and limitations.

Key words. nonexpansive operator, fixed-point problem, Fast KM, stochastic oracle, almost-sure convergence.

1 Introduction

We consider the fixed-point problem

$$\text{find } x^\star \in \mathbb{R}^d \quad \text{such that} \quad Tx^\star = x^\star, \tag{FP}$$

where $T : \mathbb{R}^d \to \mathbb{R}^d$ is a nonexpansive operator, that is, $\|Tx - Ty\| \le \|x - y\|$ for all $x, y \in \mathbb{R}^d$, whose set of fixed points $\operatorname{Fix} T := \{x \in \mathbb{R}^d : Tx = x\}$ is nonempty. Throughout, only noisy evaluations of T are available.

For exact evaluations, the fixed-point residual of KM with a constant relaxation parameter satisfies $\|x_k - Tx_k\| = o(1/\sqrt{k})$ [2]. The Fast KM method of Boţ and Nguyen [2, Algorithm 2.1] improves this to

$$\|x_k - Tx_k\| = o(1/k), \qquad x_k \to x^\star \in \operatorname{Fix} T, \tag{1.1}$$

for a momentum parameter $\alpha > 2$.

With unit stepsize, the Fast KM update reads

$$x_{k+1} = x_k - \frac{\alpha}{2(k+\alpha)}(x_k - Tx_k) + \frac{k}{k+\alpha}(Tx_k - Tx_{k-1}), \qquad k \ge 1, \tag{1.2}$$

and uses the operator values at x_k and x_{k-1}. With noisy evaluations, Tx_k is replaced by an oracle response $\widehat{T}_k(x_k) = Tx_k + e_k$, where the subscript k indicates the sample drawn at iteration k. For the difference $Tx_k - Tx_{k-1}$ there are two implementations:

- *Re-query with the same sample:* $\widehat{T}_k(x_k) - \widehat{T}_k(x_{k-1})$. Both values use the sample of iteration k. This requires two queries per iteration and an oracle that can evaluate one sample at two points [7, Assumption 2.3]. If the noise is Lipschitz in mean square with constant L_Δ [7, Assumption 2.4], the error of the difference is at most $L_\Delta\|x_k - x_{k-1}\|$ in root mean square; see Appendix A. Variance-reduced methods build on this access [7–9].
- *Reuse:* $\widehat{T}_k(x_k) - \widehat{T}_{k-1}(x_{k-1})$. The second value is the response stored at the preceding step, so each iteration queries the oracle once, as in the deterministic method. The error of the difference is $e_k - e_{k-1}$, which involves two independent samples and does not shrink as $x_k - x_{k-1} \to 0$.

We thus study the reuse implementation. Table 1 raises three questions:

1. *(Q1) Last iterate of stochastic KM.* Under bounded variance, the best available last-iterate bound is $O(\log k/k^{1/6})$, whereas a randomly selected iterate attains $O(k^{-1/4})$. Does the last iterate necessarily lose this rate? (Section 3)
2. *(Q2) Implicit variance reduction in Fast KM.* Under bounded variance, does the momentum of Algorithm 1 with fixed $\alpha > 2$ reduce the variance implicitly, so that its last iterate improves on that of stochastic KM? (Section 4)
3. *(Q3) Fast KM under decaying noise.* For $\sigma_k^2 \asymp k^{-q}$, which decay of the variance enables Algorithm 1 to attain $k\|x_k - Tx_k\| \to 0$ and $x_k \to x_\infty$ almost surely? (Section 5)

Table 1: Comparison of last-iterate residual guarantees for a nonexpansive T on $\mathbb{R}^d$.

Method	Momentum coefficient	Last-iterate rate
KM [2]	—	$o(1/\sqrt{k})$
Stochastic KM [3]	—	$O(\log k/k^{1/6})$
Fast KM [2]	$\frac{k}{k+\alpha} \to 1$	$o(1/k)$
Stochastic Fast KM, reuse (this paper)	$\frac{k}{k+\alpha} \to 1$	need not vanish in mean square (Prop. 4.1)
Same, $\mathbb{E}\sigma_k^2 \asymp k^{-q}$ (this paper)	$\frac{k}{k+\alpha} \to 1$	mean-square: $q > 2$ necessary; a.s.: open ($q > 3$ candidate)

One oracle query per iteration. Stochastic KM is stated in expectation; the fixed-parameter counterexample concerns mean-square convergence. The last row states an almost-sure research objective; its mean-square necessary condition is a separate conclusion. Proposition 4.1 assumes $\mathbb{E}\sigma_k^2 \geq \sigma^2 > 0$.

2 Stochastic Oracle Setting

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(\mathcal{F}_k)_{k\geq 0}$ a filtration, where $\mathcal{F}_k$ contains the information available before query k. At an $\mathcal{F}_k$-measurable point x_k, the oracle returns

$$\widehat{T}_k(x_k) = Tx_k + e_k \tag{2.1}$$

with an $\mathcal{F}_{k+1}$-measurable error e_k.

Assumption 2.1 (Oracle)**.** For every $k \geq 0$,

$$\mathbb{E}[e_k \mid \mathcal{F}_k] = 0 \qquad \text{and} \qquad \sigma_k^2 := \mathbb{E}\big[\|e_k\|^2 \;\big|\; \mathcal{F}_k\big] \text{ with } \mathbb{E}\sigma_k^2 < \infty. \tag{H1}$$

3 The last iterate of stochastic KM

In this section, we review existing convergence results for stochastic Krasnosel'skiĭ–Mann iterations in [3].

The stochastic KM iteration is

$$x_{k+1} = (1-\lambda_{k+1})x_k + \lambda_{k+1}\widehat{T}_k(x_k), \qquad \widehat{T}_k(x_k) = Tx_k + e_k. \tag{3.1}$$

Under bounded variance, a randomly selected iterate generated by (3.1) attains the order $k^{-1/4}$ [3, Corollary 5.4], [4]. This leads to the following question:

(Q1) For a general nonexpansive T and bounded variance, can the last iterate of stochastic KM attain the order $k^{-1/4}$?

Let $D := \operatorname{dist}(x_0, \operatorname{Fix} T)$ and set $\tau_k := \sum_{j=1}^k \lambda_j(1-\lambda_j)$, $\varphi(y) := \min\{1, 1/\sqrt{\pi y}\}$ and

$$\nu_k^2 := \sum_{j=1}^k \big(w_j^k\big)^2 \mathbb{E}\sigma_{j-1}^2, \qquad w_j^k := \lambda_j \prod_{i=j+1}^k (1-\lambda_i),$$

We recall the conditions used for the last-iterate bound in [3].

Assumption 3.1. Either T has bounded range and $\bar{\kappa} \geq \sup_x \|Tx - x_0\|$, or $S := \sum_{j\geq 2} \lambda_j \nu_{j-1} < \infty$ and $\bar{\kappa} \geq 2D + S$.

The following theorem gives a bound on the expected fixed-point residual of the last iterate of stochastic KM [3, Theorem 2.10].

Theorem 3.1. *Let T be nonexpansive with $\operatorname{Fix} T \neq \emptyset$, and let Assumptions 2.1 and 3.1 hold. Then, for every $k \geq 1$,*

$$\mathbb{E}\|x_k - Tx_k\| \leq \bar{\kappa}\,\varphi(\tau_k) + 2\sum_{j=2}^k \lambda_j\,\varphi(\tau_k - \tau_j)\,\nu_{j-1} + 4\nu_k. \tag{3.2}$$

We now consider the special case of Assumption 2.1 in which the second moments are uniformly bounded.

Algorithm 1 Stochastic Fast KM with reuse

Input: $x_0, x_1 \in \mathbb{R}^d$, $\alpha > 2$.
Query $\widehat{T}_0(x_0)$ and store it.
for $k = 1, 2, \dots$ **do**
  Query $\widehat{T}_k(x_k) = Tx_k + e_k$.
  $x_{k+1} = x_k - \frac{\rho_k}{2}\big(x_k - \widehat{T}_k(x_k)\big) + (1-\rho_k)\big(\widehat{T}_k(x_k) - \widehat{T}_{k-1}(x_{k-1})\big)$.
  Store $\widehat{T}_k(x_k)$.
end for

Assumption 3.2 (Bounded variance)**.** There exists a finite constant $\sigma \geq 0$ such that $\sup_{k\geq 0} \mathbb{E}\sigma_k^2 \leq \sigma^2$.

Corollary 3.2. *[3, Theorem 4.4] Under the assumptions of Theorem 3.1 and Assumption 3.2, let $\lambda_k = (k+1)^{-a}$ with $a \in (\frac{1}{2}, 1)$. Then $\tau_k \asymp k^{1-a}$ and it holds that*

$$\mathbb{E}\|x_k - Tx_k\| \lesssim \underbrace{\frac{1}{\sqrt{\tau_k}}}_{deterministic} + \underbrace{\sigma\sqrt{k}\,\lambda_k}_{noise} = \frac{1}{k^{(1-a)/2}} + \frac{\sigma}{k^{a-1/2}}.$$

Moreover, it holds that $\mathbb{E}\|x_k - Tx_k\| = O(\log k / k^{1/6})$ with $a = \frac{2}{3}$.

Answer to (Q1). In the literature examined here, no: for a general nonexpansive T,

$$\mathbb{E}\|x_k - Tx_k\| = O\Big(\frac{\log k}{k^{1/6}}\Big) \quad \text{(last iterate)}, \qquad \mathbb{E}\|x_R - Tx_R\| = O\Big(\frac{1}{k^{1/4}}\Big) \quad \text{(random iterate)}.$$

The limitations are as follows:

1. For exact evaluations,

$$\|x_{k+1} - Tx_{k+1}\| \leq \|x_k - Tx_k\| \quad \Longrightarrow \quad \|x_k - Tx_k\|^2 \leq \frac{D^2}{\tau_{k+1}};$$

   with noise, $\mathbb{E}\|x_k - Tx_k\|^2$ need not be monotone, and the summed bound applies only to x_R [3, Remark 5.3].

2. *Order of the last iterate:* in (3.2), $1/\sqrt{\tau_k} \asymp \sqrt{k}\,\lambda_k \asymp k^{-1/6}$ at $a = \frac{2}{3}$ (Corollary 3.2).

4 Implicit variance reduction in Fast KM

4.1 The reuse implementation of Stochastic Fast KM

The Fast KM method [2, Algorithm 2.1] also has a stepsize $0 < s \leq 1$. Its iteration with stepsize s applied to T coincides with (1.2) applied to $M := (1-s)\,\mathrm{Id} + sT$, which is nonexpansive with $\mathrm{Fix}\, M = \mathrm{Fix}\, T$, and an oracle for T with error e_k yields an oracle for M with error se_k. We therefore take $s = 1$ without loss of generality.

Set $\rho_k := \frac{\alpha}{k+\alpha}$, so that $\frac{\alpha}{2(k+\alpha)} = \frac{\rho_k}{2}$ and $\frac{k}{k+\alpha} = 1 - \rho_k$ in (1.2). Algorithm 1 replaces Tx_k by the fresh response $\widehat{T}_k(x_k) = Tx_k + e_k$ and Tx_{k-1} by the stored response $\widehat{T}_{k-1}(x_{k-1}) = Tx_{k-1} + e_{k-1}$.

In stochastic optimization, momentum averages the sampled gradients and reduces the variance implicitly [5, 6]. This leads to the second question:

(Q2) Under bounded variance, does the momentum of Algorithm 1 with fixed $\alpha > 2$ reduce the variance implicitly, so that its last iterate improves on that of stochastic KM?

4.2 Noise propagation under reuse

The update of Algorithm 1 splits into the exact step and the errors:

$$x_{k+1} = \underbrace{x_k - \frac{\rho_k}{2}(x_k - Tx_k) + (1-\rho_k)(Tx_k - Tx_{k-1})}_{\text{exact Fast KM step}} + \underbrace{\frac{\rho_k}{2}}_{\text{relaxation}\to 0} e_k + \underbrace{(1-\rho_k)}_{\text{momentum}\to 1}(e_k - e_{k-1}).$$

Compared with (1.2), the update thus carries the error

$$\eta_k = \Big(1 - \frac{\rho_k}{2}\Big) e_k - (1 - \rho_k)\, e_{k-1}. \tag{4.1}$$

The stored error e_{k-1} enters η_{k-1} and η_k with coefficients of opposite signs, which nearly cancel (Lemma 4.2). The new error e_k enters with the weight

$$\underbrace{\frac{\rho_k}{2}}_{\text{relaxation}} + \underbrace{(1 - \rho_k)}_{\text{momentum}} = 1 - \frac{\rho_k}{2} \;\to\; 1,$$

so the momentum term offsets the damping of the relaxation step. For $T = 0$, this gives a lower bound on the residual.

Proposition 4.1. *Let $T = 0$ and assume $\mathbb{E}\sigma_k^2 \geq \sigma^2 > 0$ for all k. Then the iterates of Algorithm 1 satisfy, for every $k \geq 1$,*

$$\mathbb{E}\|x_{k+1} - Tx_{k+1}\|^2 \geq \frac{\sigma^2}{4}.$$

Proof. For $T = 0$, the update reads $x_{k+1} = \big(1 - \frac{\rho_k}{2}\big)x_k + \big(1 - \frac{\rho_k}{2}\big)e_k - (1 - \rho_k)e_{k-1}$. Since x_k and e_{k-1} are $\mathcal{F}_k$-measurable, conditional unbiasedness gives $\mathbb{E}\|x_{k+1}\|^2 \geq \big(1 - \frac{\rho_k}{2}\big)^2 \mathbb{E}\sigma_k^2 \geq \frac{\sigma^2}{4}$, because $\rho_k \leq 1$. □

Answer to (Q2). For fixed $\alpha > 2$: no. The residual of Algorithm 1 need not even vanish in mean square under bounded variance (Proposition 4.1).

The perturbation argument. Existing analyses of Fast KM concern exact evaluations or summable perturbations [10], and bounded variance does not make η_k in (4.1) summable. Following [3, Section 2.1], we reformulate the iteration with an auxiliary sequence that absorbs the errors. Let $u_0 := 0$, $u_1 := \frac{1}{2}e_0$, and define

$$u_{k+1} = \Big(1 - \frac{\rho_k}{2}\Big)u_k + \Big(1 - \frac{\rho_k}{2}\Big)e_k - (1 - \rho_k)\, e_{k-1}, \qquad k \geq 1. \tag{4.2}$$

Subtracting this from the update and denoting $z_k := x_k - u_k$, we get

$$z_{k+1} = z_k - \frac{\rho_k}{2}\big(z_k - Tz_k - \delta_k\big) + (1 - \rho_k)\big(Tz_k + \delta_k - Tz_{k-1} - \delta_{k-1}\big),$$

which is the deterministic Fast KM iteration with errors $\delta_k := Tx_k - Tz_k$ satisfying $\|\delta_k\| \leq \|x_k - z_k\| = \|u_k\|$.

For stochastic KM, the corresponding process is an average of the errors and converges to zero under the noise and stepsize conditions in that analysis, so the effective errors δ_k become small [3, Section 2.1]. The next lemma shows that u_k is not an average.

Lemma 4.2 (Noise propagation under reuse)**.** *Let Assumption 2.1 hold. Set $v_k := u_k - \big(1 - \frac{\rho_{k-1}}{2}\big)e_{k-1}$ and $b_k := \big(1 - \frac{\rho_k}{2}\big)\big(1 - \frac{\rho_{k-1}}{2}\big) - (1 - \rho_k)$. Then $v_1 = 0$, $0 \leq b_k \leq \frac{\rho_k \rho_{k-1}}{4}$, and for all $k \geq 1$,*

$$u_{k+1} = \Big(1 - \frac{\rho_k}{2}\Big)e_k + v_{k+1}, \qquad v_{k+1} = \Big(1 - \frac{\rho_k}{2}\Big)v_k + b_k\, e_{k-1}, \tag{4.3}$$

$$\begin{aligned} \mathbb{E}\|u_{k+1}\|^2 &= \Big(1 - \frac{\rho_k}{2}\Big)^2 \mathbb{E}\sigma_k^2 + \mathbb{E}\|v_{k+1}\|^2, \\ \mathbb{E}\|v_{k+1}\|^2 &= \Big(1 - \frac{\rho_k}{2}\Big)^2 \mathbb{E}\|v_k\|^2 + b_k^2\, \mathbb{E}\sigma_{k-1}^2. \end{aligned} \tag{4.4}$$

Proof. Since $\rho_0 = 1$, $v_1 = u_1 - \frac{1}{2}e_0 = 0$. Substituting $u_k = v_k + \big(1 - \frac{\rho_{k-1}}{2}\big)e_{k-1}$ into the recursion of u_k gives the recursion of v_k. A direct computation gives $b_k = \frac{\alpha(\alpha-2)}{4(k+\alpha)(k+\alpha-1)}$, which lies in $\big[0, \frac{\rho_k \rho_{k-1}}{4}\big]$. The vector v_k depends only on $e_0, \ldots, e_{k-2}$, and v_{k+1} is $\mathcal{F}_k$-measurable. Conditional unbiasedness therefore removes the cross terms. □

Remark 4.3. By Lemma 4.2, the stored error enters u_{k+1} with the net coefficient $b_k = O(1/k^2)$ and the new error with the weight $1 - \frac{\rho_k}{2} \to 1$. Hence, if $\inf_k \mathbb{E}\sigma_k^2 > 0$, then u_k does not converge to zero in mean square, and the estimate $\|\delta_k\| \leq \|u_k\|$ does not make the effective errors small. Bravo and Cominetti observe that stochastic KM carries implicitly a form of variance reduction that operates as the iterates evolve [3, Remark 2.3]. For Algorithm 1, this variance reduction acts only on the stored errors. Momentum-based methods control the contribution of new stochastic information through tunable parameters [5, 6].

5 Fast KM under decaying noise

Proposition 4.1 shows that the noise has to decay. Its proof gives, for $T = 0$,

$$\mathbb{E}\|x_{k+1} - Tx_{k+1}\|^2 \geq \Big(1 - \frac{\rho_k}{2}\Big)^2 \mathbb{E}\sigma_k^2 \geq \frac{1}{4}\mathbb{E}\sigma_k^2,$$

so the root-mean-square rate $(\mathbb{E}\|x_k - Tx_k\|^2)^{1/2} = o(1/k)$ for $T = 0$ requires

$$k^2\,\mathbb{E}\sigma_k^2 \longrightarrow 0. \tag{5.1}$$

A candidate sufficient condition for almost-sure convergence and rates is the decaying-noise condition $\sum_k s_k^2 t_k^2 \sigma_k^2 < \infty$ almost surely of [1, Theorem 3.1], where s_k is the stepsize and $t_k := 1 + \sum_{i \geq k} \prod_{j=k}^{i} \alpha_j$ for the extrapolation coefficients α_k. For Algorithm 1, $s_k \equiv 1$ and $\alpha_k = 1 - \rho_k$, so $t_k = \frac{k+\alpha-1}{\alpha-1}$ and the condition reads

$$\sum_k (k+1)^2 \sigma_k^2 < \infty \qquad \text{almost surely.} \tag{N6-ℓ^2}$$

Theorem 3.1 of [1] is stated for the Ravine method, and its applicability to Algorithm 1 remains to be established.

Both conditions concern the weighted variance $k^2\sigma_k^2$: (5.1) requires it to vanish in expectation, and (N6-ℓ^2) requires it to be summable. For $\sigma_k^2 \asymp k^{-q}$, they correspond to $q > 2$ and $q > 3$, respectively. The former is necessary for the mean-square rate; the latter is a candidate condition for the almost-sure rate. This leads to the third question:

> *(Q3) For $\sigma_k^2 \asymp k^{-q}$, which decay ensures that Algorithm 1 attains $k\|x_k - Tx_k\| \to 0$ and $x_k \to x_\infty$ almost surely, for a $\operatorname{Fix} T$-valued random variable x_∞? In particular, does the almost-sure threshold lie at $q > 2$, as for the mean-square rate, or at $q > 3$?*

A Re-query with the same sample

This appendix compares the two implementations of Section 1 in the notation of Section 4. Let $T(\cdot;\xi)$ be a sampled operator with $\mathbb{E}[T(y;\xi_k)] = Ty$ for every $y \in \mathbb{R}^d$, where the sample ξ_k is independent of $\mathcal{F}_k$. Set $\widehat{T}_k(y) := T(y;\xi_k)$ and

$$e_k(y) := \widehat{T}_k(y) - Ty, \qquad y \in \mathbb{R}^d.$$

The error of Section 4 is $e_k = e_k(x_k)$. Since x_k and x_{k-1} are $\mathcal{F}_k$-measurable, $\mathbb{E}[e_k(x_k) \mid \mathcal{F}_k] = \mathbb{E}[e_k(x_{k-1}) \mid \mathcal{F}_k] = 0$.

Two updates. With $\rho_k = \frac{\alpha}{k+\alpha}$, the re-query implementation evaluates $\widehat{T}_k$ at x_k and at x_{k-1}:

$$x_{k+1} = x_k - \frac{\rho_k}{2}\big(x_k - \widehat{T}_k(x_k)\big) + (1-\rho_k)\big(\widehat{T}_k(x_k) - \widehat{T}_k(x_{k-1})\big). \tag{A.1}$$

This requires two queries per iteration and same-sample access [7, Assumption 2.3]. Algorithm 1 replaces $\widehat{T}_k(x_{k-1})$ by the stored response $\widehat{T}_{k-1}(x_{k-1})$ and uses each sample once.

Errors. Compared with (1.2), the two updates carry the errors

$$\eta_k^{\mathrm{rq}} = \frac{\rho_k}{2}\,e_k + (1-\rho_k)\big(e_k - e_k(x_{k-1})\big), \qquad \eta_k = \frac{\rho_k}{2}\,e_k + (1-\rho_k)\big(e_k - e_{k-1}\big). \tag{A.2}$$

They differ only in the error at x_{k-1}: the re-query implementation uses the error $e_k(x_{k-1})$ of the current sample, and the reuse implementation uses the error $e_{k-1} = e_{k-1}(x_{k-1})$ of the preceding sample.

Assume the noise is Lipschitz in mean square, as in [7, Assumption 2.4]: $\mathbb{E}\|e_k(x) - e_k(y)\|^2 \leq L_\Delta^2\|x-y\|^2$ for all $x, y \in \mathbb{R}^d$. Since x_k and x_{k-1} are $\mathcal{F}_k$-measurable and ξ_k is independent of $\mathcal{F}_k$,

$$\mathbb{E}\big[\|e_k(x_k) - e_k(x_{k-1})\|^2 \;\big|\; \mathcal{F}_k\big] \leq L_\Delta^2\|x_k - x_{k-1}\|^2.$$

In the reuse implementation, e_{k-1} is $\mathcal{F}_k$-measurable and $\mathbb{E}[e_k \mid \mathcal{F}_k] = 0$, so

$$\mathbb{E}\|e_k - e_{k-1}\|^2 = \mathbb{E}\sigma_k^2 + \mathbb{E}\sigma_{k-1}^2,$$

which does not decrease as $x_k - x_{k-1} \to 0$. Hence, in η_k^{rq} the new error e_k enters outside the difference only with the weight $\frac{\rho_k}{2}$, whereas in η_k it enters with the weight $1 - \frac{\rho_k}{2}$ (Lemma 4.2).

Variance-reduced estimators. The STORM-type recursion of VRAF [7, (VRAF)], written for T in the present indexing, is

$$d_k = \widehat{T}_k(x_k) + (1-\beta_k)\big(d_{k-1} - \widehat{T}_k(x_{k-1})\big), \qquad \beta_k \in (0,1].$$

Its error $\varepsilon_k := d_k - Tx_k$ satisfies

$$\varepsilon_k = (1-\beta_k)\,\varepsilon_{k-1} + \beta_k\, e_k + (1-\beta_k)\big(e_k - e_k(x_{k-1})\big).$$

The new error enters with the weight β_k, and the remaining term is the same-sample difference controlled above. The loopless SVRG estimator of [8, (7)] for Fast KM likewise evaluates the same mini-batch at x_k and at x_{k-1}. Both constructions rely on same-sample access, which the reuse implementation does not assume.

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